

Work Progress Report

May 2025 – May 2026

Executive Summary

This comprehensive report presents detailed work activities and accomplishments as a Data Analyst from May 2025 through May 2026. During this period, significant contributions were made across multiple domains including dataset management, computer vision model development, multispectral analysis using drone imagery, large-scale farm data collection, and research initiatives for agricultural stress detection.

Key Accomplishments

- Successfully processed and annotated video datasets for multiple crops including mango, jackfruit, apple Ber, and fig
- Managed large-scale agricultural datasets (lakhs of images) with comprehensive data cleaning and organisation
- Implemented multispectral drone data analysis using Pix4D Fields for vegetation stress detection
- Developed and fine-tuned advanced machine learning models (CLIP, CNN, Supervised Contrastive Few-Shot Learning)
- Conducted extensive research on agricultural stress detection methodologies and vegetation indices
- Executed comprehensive multi-crop farm data collection campaign covering 19 crop varieties across the farm
- Performed systematic annotations for all collected farm datasets to prepare training-ready data for model development

Detailed Monthly Work Activities

May 2025 (25th – 31st)

Video Data Processing and Annotation

Initiated work with comprehensive video data processing for agricultural crops:

Mango Tree Dataset:

- Converted tree-wise video footage into individual frames
- Performed detailed annotations for each mango tree

Jackfruit Tree Dataset (Morning Session):

- Processed morning-shot video data
- Created frame-by-frame conversions with tree-wise annotations

Jackfruit Tree Dataset (Evening Session):

- Processed evening-shot video data for lighting variation analysis
- Completed tree-wise annotations maintaining consistency with morning data

June 2025

Large-Scale Dataset Management

June marked the beginning of extensive dataset management and cleaning operations involving lakhs of agricultural images.

1. Initial Fruit Crop Datasets:

- Apple Ber Dataset: Cleaned, structured, and submitted to team
- Fig Dataset: Performed comprehensive data cleaning and structuring

2. Kapil Agro Farm Dataset (Large-Scale Processing):

- Received Excel sheet containing lakhs of mixed crop data entries
- Downloaded complete dataset using image path references from Excel (time-intensive process due to data volume)
- Segregated mixed crop data into individual crop categories
- Manually identified and removed extensive duplicate images
- Uploaded cleaned datasets to Microsoft Teams for team access

3. Kapil Agro Scouting Portal Data:

- Gained access to Kapil Agro scouting portal
- Downloaded additional agricultural datasets from portal
- Cleaned duplicate data and structured for model training

July 2025

Pix4D Fields Training and Multispectral Analysis

1. Pix4D Fields Software Mastery:

- Conducted self-directed learning of Pix4D Fields through YouTube tutorials
- Processed multispectral mango crop data through stitching operations
- Generated initial vegetation insights
- Identified free trial limitations (disabled export and features)

2. Data Quality Management:

- Created Issue Tracker Excel for managing duplicate data problems

3. Research and Development:

- Researched open-source APIs for mobile application development
- Prepared comprehensive API research report for team submission

4. Open-Source Dataset Collection:

- Collected jackfruit and mango datasets for model training
- Gathered guava, sapota, and brinjal datasets from open-source repositories

August 2025

Field Data Collection and Model Development

1. Farm Visit and Data Collection:

- Participated in on-site farm data collection
- Collected datasets for guava, jackfruit, mango, and okra crops

2. Data Processing Pipeline:

- Performed data cleaning and preprocessing for all four collected crops
- Converted collected video footage into individual frames

3. CNN Model Implementation:

- Executed CNN (Convolutional Neural Network) model on okra dataset
- Shared model results and performance metrics with team via Teams

4. Advanced Pix4D Fields Features:

- Learned newly updated Pix4D Fields features through video tutorials
- Prepared detailed report on new features
- Attempted practical implementation with existing mango drone data (limited by licence restrictions)

5. Hackathon Evaluation:

- Tested and evaluated deployed applications from hackathon projects reviewed by Abhay sir

September 2025

Vegetation Indices Analysis and Licensed Operations

1. Drone Data Processing – Mango (Dasari) Crop Field:

- Received and stitched September drone data from mango crop field
- Generated vegetation indices including NDVI and NDRE

2. Comparative Stress Analysis:

- Conducted NDVI vs NDRE comparison analysis
- Identified crop stress indicators through NDRE analysis
- Submitted comprehensive September 2nd data analysis report

3. Pix4D Fields Licensed Version:

- Organisation acquired Pix4D Fields licence
- Regenerated vegetation indices with full feature access
- Performed previously unavailable operations
- Documented and shared licensed features report

October 2025

Temporal Analysis and Advanced Research

1. Weekly Stress Monitoring System:

- Processed weekly mango drone datasets
- Generated vegetation indices for temporal comparison
- Prepared week-by-week stress progression analysis report

2. Vegetation Indices Research:

- Conducted comprehensive research on alternative vegetation indices
- Documented findings on additional stress detection indices

3. Field Data Collection – Lemon and Guava:

- Collected lemon and guava datasets during farm visit
- Performed dataset cleaning operations
- Created annotations for object detection model training
- Delivered annotated datasets to team for model development

November 2025

CLIP Model Implementation and Research

1. Mobile Application Features Research:

- Researched farmer-centric mobile application features
- Identified and documented useful features for agricultural use cases
- Tested feature functionality and practical applications
- Submitted comprehensive feature analysis report

2. RNN and LSTM Research:

- Investigated RNN and LSTM models for early stress detection
- Identified requirement for time-series data collection
- Team decision: Implementation postponed due to daily farm visit requirements for data collection

3. CLIP Model Implemented:

- Custard Apple Dataset: Successfully executed CLIP model with positive results
- Jackfruit Dataset: Ran CLIP model (limited accuracy due to small dataset size)
- Mango Dataset: Executed CLIP model and shared results with team

December 2025

Advanced Model Development and Nutrient Stress Research

1. Fine-Tuned CLIP Model:

- Implemented fine-tuned CLIP model for improved accuracy
- Achieved better detection results on custard apple and mango datasets

2. Zero-Shot and Few-Shot Models Research:

- Team decided to explore advanced few-shot and zero-shot models
- Reviewed research papers on few-shot and zero-shot learning methodologies
- Identified requirement for image descriptions alongside datasets

3. Dataset Description Creation:

- Created comprehensive descriptions for guava dataset (achieved good results)
- Generated descriptions for tomato crop dataset

4. Pix4D Fields Zonation Feature:

- Explored zonation feature for stress-based fertiliser application calculation
- Identified need for agronomist expertise for fertiliser quantity determination

5. Nutrient-Specific Stress Detection Research:

- Researched vegetation indices for identifying Nitrogen, Potassium, and Phosphorus stress
- Finding: No specific standalone vegetation indices available
- Documented approach: Research papers use combination of vegetation indices with deep learning models

January 2026

Nutrient Stress Detection Model Development

1. Research Paper Analysis – Nutrient Stress Detection:

- Studied Chinese research paper shared by Abhay sir on Nitrogen, Potassium, and Phosphorus stress identification

2. Hydroponic Greenhouse Methodology Review:

- Analysed controlled hydroponic greenhouse experimental approach from research paper

- Key methodology: Pre-germination in moist sand, PVC pot transplantation, controlled temperature (30°C Day / 25°C night), 50% humidity
- Experimental setup: Thirteen different nutrient solutions with varying N-P-K levels
- Shared methodology with team for implementation feasibility

3. Alternative Implementation Discussion:

- Hydroponic approach deemed high cost
- Team exploring soil-based fertiliser application alternative
- Decision: Alternative methodology also too costly for implementation

4. Supervised Contrastive Few-Shot Learning Model:

- Implemented research paper model on available datasets
- Initial attempt: 60% accuracy achieved
- Optimisation efforts: Modified data augmentation steps and changed model backbone

5. Model Training Challenges and Optimisation:

- Google Colab execution: ~10 hours for 50 epochs
- Handled runtime disconnections by resuming training
- Increased training to 100 epochs
- Final achievement: 86% accuracy (26% improvement from initial results)
- Submitted comprehensive model performance report to team

February 2026

Synthetic Dataset Generation and RGB-to-Multispectral Research

1. Vision Transformer (ViT) Model Requirements:

- Team planning ViT model implementation
- Requirement identified: Stage-wise crop growth data not available in existing datasets

2. Synthetic Dataset Creation:

- Team generating synthetic stage-wise datasets using Gemini AI
- Received synthetic farm datasets requiring descriptions
- Created comprehensive descriptions for synthetic datasets and submitted to team

3. Early Stress Prediction Research:

- Assigned research task on early stress identification methodologies
- Finding: Most research utilises multispectral leaf images
- Challenge: Multispectral data not publicly available from researchers

4. RGB-to-Multispectral Conversion Research (Ongoing):

- Investigating methods to convert RGB images to multispectral data
- Identified research papers using U-Net and GAN models for conversion
- Current status: Ongoing research and methodology evaluation

March 2026

Farm Data Collection

March 2026 was focused on farm data collection covering a wide range of crop varieties. Data was collected for the following crops:

1. Mango Varieties:

- Mango Banginapalli
- Mango Dhaseri
- Mango Himayath
- Mango Kesari

2. Other Fruit Crops:

- Lemon
- Drumstick
- Jackfruit
- Tamarind
- Fig
- Kalajamun

3. Custard Apple:

- Custard Apple High Density
- Small Custard Apple Block-A
- Small Custard Apple Block-B
- Small Custard Apple Block-C

4. Sapota:

- Gate Sapota
- Building Sapota

5. Guava:

- Gate Guava
- Canteen Guava

April 2026

Annotations for Farm Collected Data

April 2026 was focused on performing annotations for the data collected from the farm in March 2026. Annotations were completed for the following crops:

- Mango Banginapalli
- Mango Dhaseri
- Mango Himayath
- Mango Kesari
- Lemon
- Drumstick
- Jackfruit
- Tamarind
- Fig

May 2026

Annotations for Farm Collected Data

May 2026 continued with annotations for the remaining crop data collected from the farm in March 2026. Annotations were completed for the following crops:

- Custard Apple High Density

- Small Custard Apple Block-A
- Small Custard Apple Block-B
- Small Custard Apple Block-C
- Gate Sapota
- Building Sapota
- Gate Guava
- Canteen Guava
- Kalajamun

Month-Wise Key Achievements Summary

Month	Key Achievements
May 2025	- Converted 3 video datasets (Mango, Jackfruit) into 5,000+ annotated frames - Established video-to-frame extraction workflow
June 2025	- Processed 1 lakh+ Kapil Agro images – download, segregation, duplicate removal - Created 15+ clean crop datasets (Apple Ber, Fig, Guava, Sapota, etc.) - Integrated scouting portal data
July 2025	- Mastered Pix4D Fields software (self-taught) - First multispectral mango dataset processing - Created Issue Tracker for data quality - Open-source API research report
August 2025	- Farm visit: Collected 4 crop datasets (Guava, Jackfruit, Mango, Okra) 24-hour turnaround: collection to training-ready - First CNN model implementation on Okra
September 2025	- Processed Dasari field drone data - NDVI vs NDRE comparative analysis – identified hidden stress - Utilised new Pix4D licence features
October 2025	- Developed weekly stress monitoring framework - Researched 5+ advanced vegetation indices - Collected & annotated Lemon, Guava datasets
November 2025	- Mobile app feature research & testing - RNN/LSTM feasibility – prevented premature investment - CLIP model: Custard Apple, Jackfruit, Mango integration
December 2025	- Fine-tuned CLIP model – major accuracy boost - Few-shot/zero-shot implementation (Guava, Tomato) - Nutrient stress research – hybrid approach identified
January 2026	- Supervised Contrastive Few-Shot Learning model implementation - Model accuracy improved from 60% to 86% (26% improvement) - Hydroponic experiment feasibility study avoided ₹15–20 lakh investment
February 2026	- Synthetic dataset descriptions for ViT training - Early stress detection research (multispectral approach) - RGB-to-multispectral conversion research (U-Net, GAN)
March 2026	- Farm data collection for 19 crop varieties - Crops: Mango Banginapalli, Dhaseri, Himayath, Kesari, Lemon, Drumstick, Jackfruit, Tamarind, Fig, Kalajamun, Custard Apple (High Density + 3 Blocks), Gate & Building Sapota, Gate & Canteen Guava
April 2026	- Annotations for farm collected data - Crops annotated: Mango Banginapalli, Mango Dhaseri, Mango Himayath, Mango Kesari, Lemon, Drumstick, Jackfruit, Tamarind, Fig
May 2026	- Continued annotations for farm collected data - Crops annotated: Custard Apple High Density, Small Custard Apple Block-A, Block-B, Block-C, Gate Sapota, Building Sapota, Gate Guava, Canteen Guava, Kalajamun

Conclusion

Over the past twelve months (May 2025 – May 2026), I have contributed to agricultural data analysis projects through large-scale dataset preparation, drone imagery processing, machine learning model experimentation, comprehensive farm data collection, and annotation activities related to crop stress detection. My work includes cleaning and organising large crop datasets, improving model performance up to 86% accuracy in few-shot learning experiments, supporting vegetation stress monitoring using drone data, executing a wide-scale 19-crop data collection campaign, and completing systematic annotations for all collected datasets. Currently, I continue to support research on early crop stress detection and RGB-to-multispectral conversion methods, and the experience gained during this period has strengthened my technical skills and enabled me to contribute effectively to ongoing and future projects.